University of Glasgow

SCHOOL OF MATHEMATICS AND STATISTICS

LEVEL-4 HONOURS PROJECT

Overtwisted Contact Structures and their Supporting Open Books on 3-Manifolds

Author:
Finlay Scheib
2653836S

Supervisor:
Dr Andy Wand

Declaration of Originality
I confirm that this assignment is my own work and that I have:

- *Read and understood the guidance on plagiarism in the Student Handbook, including the University of Glasgow Statement on Plagiarism*
- *Not made use of the work of any other student(s) past or present without acknowledgment. This includes any of my own work, that has been previously, or concurrently, submitted for assessment, either at this or any other educational institution, including school*
- *Not sought or used the services of any professional agencies to produce this work*
- *In addition, I understand that any false claim in respect of this work will result in disciplinary action in accordance with University regulations*
- *I am aware of and understand the University's policy on plagiarism and I certify that this assignment is my own work, except where indicated by referencing, and that I have followed the good academic practices noted above.*

Friday 7th March, 2025

Abstract

This dissertation examines the one-to-one correspondence between two classes of geometric objects, namely open book decompositions and contact structures, known as the Giroux correspondence. The primary aim of this dissertation is to investigate the nature of this correspondence, and in particular prove a theorem stating that overtwisted contact structures correspond to open books obtained through a negative stabilisation. To achieve this goal, we explore a variety of mathematical topics along the way, including manifolds, knots, and the homotopy of plane fields, to construct tools and machinery to prove this statement.

Contents

1 Introduction

In 2001 the French mathematician Emmanuel Giroux, who was blind from the age of eleven, completed his proof of a one-to-one correspondence between *open books* and *contact structures*, seemingly unrelated geometric objects one can construct on manifolds. Open book decompositions (OBD) are objects that, as the name suggests, resemble a book which has had its front and back covers glued together, letting pages extrude out from the spine. The term *open book* was first coined in a 1973 paper by H. E. Winkelnkemper [1], but an existence proof dates back to 1920 completed by J. Alexander in [2], which involves surgery on knots and is heavily topological in nature. This is in contrast to contact structures, defined using the classical notion of *differential forms*, which are more geometric in nature. This is no surprise, as contact geometry originates from the world of classical mechanics. The real world applications of contact geometry are certainly not numerous, but it does have applications in geometric optics, namely Huygens' principle, and has even been used to describe the visual cortex [3].

The statement of Giroux's correspondence involves some extra conditions to ensure uniqueness, namely *positive stabilisation* and *isotopy*. Stabilisation of open books is discussed in Section 3.3, while isotopy of contact structures means that they are the same up to a bit of wiggling allowed. The statement for 3-manifolds in all its glory is:

Theorem 1.1 (Giroux correspondence, [4])**.** *Given a closed, oriented 3-manifold M, there is a one-to-one correspondence between the following:*

$$\{\textit{Open book decompositions up to positive stabilisation on } M\}$$

and

$$\{\textit{Contact structures up to isotopy on } M\}.$$

This project is interested in the *nature* of this correspondence and in particular asking questions along the lines of: are there any rules along which certain types of open books correspond to certain types of contact structures?
We will take a particular interest in what are known as *overtwisted* contact structures, those that possess the most flexibility, in contrast to *tight* contact structures which possess some rigidity. In 1989, Russian mathematician Yakov Eliashberg was studying contact structures, particularly the dichotomy between overtwisted and tight contact structures, and set about trying to classify them (up to isotopy). He was successful in reducing the classification of overtwisted contact structures to simple homotopy theory of 2-plane fields in an outstanding paper [5]. This initially appeared to be quite a hopeless task and yet reduced to "trivial" homotopy theory.
In conjunction with the Giroux correspondence, one can ask if the classes of overtwisted contact structures correspond nicely to classes of open books? The answer is in fact yes, and is the aim of this project to prove the following statement:

A contact structure is overtwisted $\iff$ *it is supported by an open book with a negative stabilisation.*

To get there, we have of course a little work to do. In Chapter 2 we present a little manifold theory involving: definitions of manifolds, tangent spaces and regular values of maps. We conclude this chapter with a discussion on forms, necessary for defining contact structures. In Chapter 3, we define open book decompositions, provide some examples, and give an existence proof which requires a dive into knots and surgery. Finally, we discuss stabilisation and the pivotal Dehn twists. In Chapter 4, we turn our attention to contact structures. Here we provide examples, discuss the overtwisted vs tight dichotomy and show how one makes *any* contact structure overtwisted via the Lutz twist. In Chapter 5, we begin discussing 2-plane fields on 3-manifolds, where we focus on the Thom-Pontryagin construction before defining obstruction classes. In Chapter 6 we discuss Eliashberg's classification of overtwisted contact structures before proving our final Theorem 7.5 in Chapter 7. The intended readership of this project is a Level 4 maths student at the University of Glasgow.

Unless otherwise stated, we assume our manifolds M to be closed, oriented 3 dimensional manifolds.

2 Manifold Theory

The notion of a manifold is a very natural idea in mathematics and one which is used extensively in many areas of maths. Here it will form the basis on which we define more complex structures like *open books* and *contact structures*. Examples of manifolds include the unit circle S^1, the unit sphere S^2, and the torus $T^2 = S^1 \times S^1$; the latter two forming part of the class of *surfaces*, which are the class of objects that locally look like a copy of $\mathbb{R}^2$. The notion of a manifold generalises this idea to any dimension, where an n-manifold is a space which locally looks like $\mathbb{R}^n$. We provide a rigorous definition below.
Before this however, we must discuss maps between manifolds. When we discuss maps between topological spaces, we like them to be continuous, with the usual definition. However, between manifolds we like a somewhat stronger notion, namely the notion of maps being *smooth*.

Definition 2.1. [6] Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$, then $f : U \to V$ is *smooth* if its partial derivatives of all orders exist.

From this definition, we have a one-way implication that smooth maps are necessarily continuous. If a smooth map $f : U \to V$ is also a bijection and its inverse, f^{-1}, is also smooth, we call such a map f a *diffeomorphism*. Notice how we have picked the domain and co-domain to be living in some ambient euclidean space, which is required to define the notion of partial derivatives. Whilst most manifolds we will deal with will, or could, embed in some larger euclidean space, we will define our manifolds completely abstractly without them living in some larger space. However, we still wish to talk about differentials on our manifolds, hence we must exploit the local euclidean behaviour of the manifold. We do this via the notion of *coordinate charts* (sometimes called just charts) which are maps from neighbourhoods of our manifold into open balls in $\mathbb{R}^n$. With this in mind we define a smooth manifold:

Definition 2.2. [7] Let $n \in \mathbb{N}_0$. A *smooth n-manifold* is a (second-countable, Hausdorff) topological space M, with each $x_i \in M$ contained in an open neighbourhood $U_i \subset M$ that maps homeomorphically onto an open ball $B_i \subseteq \mathbb{R}^n$, by $\phi_i : U_i \to B_i$, such that for $i, j \in I$, the *transition map*, as seen below, is smooth

$$\phi_{ij} := \phi_i \circ \phi_j^{-1} : \phi_j(U_j \cap U_i) \to \phi_i(U_j \cap U_i). \tag{2.1}$$

The collection $\{U_i\}_{i\in I}$, for some indexing set I, forms an open cover of the manifold M, and together with the coordinate charts ϕ_i forms what is known as an *atlas*, $\mathcal{A} := \{U_i, \phi_i\}_{i\in I}$. This set-up with coordinate charts is extremely useful, as we can now have manifolds that do not live in some larger ambient euclidean space, yet we will still be able to discuss concepts like bases and differentiation by travelling between our manifold and some $\mathbb{R}^n$ along these charts. The requirement for the transition map to be smooth ensures that when different charts $(U_i, \phi_i), (U_j, \phi_j)$ in the atlas overlap that they agree on such an intersection; which would be a serious problem if this was not the case. An illustration of such a scenario can be seen in Figure 1.

Example 2.3 (Coordinate charts in action, [6])**.** Consider the two-sphere, $S^2 = \{(x, y, z) \in \mathbb{R}^3 \,|\, x^2 + y^2 + z^2 = 1\} \subseteq \mathbb{R}^3$. Let $n = (0, 0, 1)$ be the North pole and $s = (0, 0, -1)$ the South pole. We choose

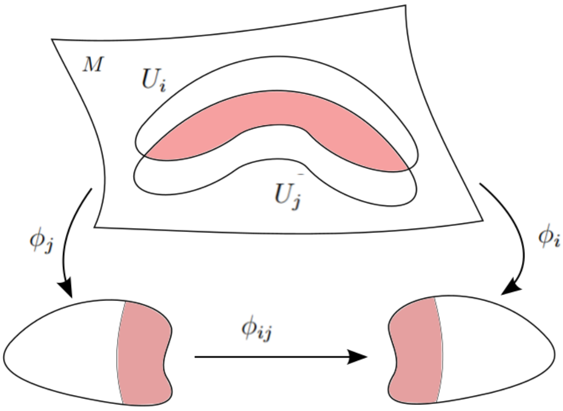

Figure 1: Charts $(U_i, \phi_i), (U_j, \phi_j)$ on a manifold M, on whose intersection we have the transition map; required to be smooth.

open sets of S^2 that will map into $\mathbb{R}^2$. We can do this using just two open sets and two charts namely,

$$\begin{aligned} &U_1 = S^2 \setminus \{n\} \text{ and chart } \phi_1 : U_1 \to \mathbb{R}^2 \text{ given by } (x, y, z) \mapsto (\frac{x}{1-z}, \frac{y}{1-z}), \\ &U_2 = S^2 \setminus \{s\} \text{ and chart } \phi_2 : U_2 \to \mathbb{R}^2 \text{ given by } (x, y, z) \mapsto (\frac{x}{1+z}, \frac{y}{1+z}). \end{aligned} \tag{2.2}$$

Since $U_1 \cup U_2 = S^2 \setminus \{n\} \cup S^2 \setminus \{s\} = S^2$, the collection of open sets cover the manifold S^2, as required. Along with our charts ϕ_1, ϕ_2 we can construct an atlas $\mathcal{A} = \{U_i, \phi_i\}_{i=1,2}$ for the manifold S^2. This example is known in the literature as *stereographic projection*, where we are mapping the sphere, with a point removed, onto the x-y plane; a copy of $\mathbb{R}^2$.

We have claimed that our manifolds need not live in some ambient euclidean space to discuss derivatives of functions. For this we need to discuss *tangent spaces*; vector spaces themselves, which form the domain and co-domain of these derivatives.

2.1 Tangent Spaces

Informally, the tangent space of an n-manifold M at a point $x \in M$, denoted T_xM, is the generalisation of a tangent line to a curve or a tangent plane to a surface. It is itself a vector space with dimension equivalent to that of the manifold itself. These qualities are precisely what we seek for the domain and co-domain of a derivative; derivatives are always tangential to their functions.
We formally define the notion of a tangent space through the idea of tangent curves. Let $x \in M$ and let $U \subseteq M$ be an open neighbourhood of x. We then make a choice of coordinate chart $\phi : U \to \mathbb{R}^n$. We then consider the following set of tangent curves that are *initialised* at $x \in M$, $\mathfrak{C} = \{c : (-1, 1) \to M \mid c(0) = x\}$, before defining an equivalence relation on this set by:

$$c_1 \sim c_2 \iff (\phi \circ c_1)'(0) = (\phi \circ c_2)'(0) \text{ for } c_1, c_2 \in \mathfrak{C}. \tag{2.3}$$

Where, $\phi \circ c_1, \phi \circ c_2 : (-1, 1) \to \mathbb{R}^n$ are differentiated as we would any other function between open euclidean subsets. The equivalence classes generated from this relation are the tangent vectors and the set of all these equivalence classes is precisely our tangent space, at x, T_xM. The tangent space is itself is a linear vector space, of equivalent dimension to the manifold itself. As with any linear vector space, we can define a notion of its *dual* space, i.e. the set of all linear maps on the tangent space to the base field (in our case $\mathbb{R}$). This is called the co-tangent space, denoted T_x^*M.

Definition 2.4. [7] Let M be an n-manifold and $x \in M$. Then the *co-tangent space* of M at x is,

$$T_x^*M = \{\omega : T_xM \to \mathbb{R} \mid \omega \text{ is linear }\}.$$

We note that, up until now, the tangent spaces have been dependent on the choice of point in the manifold. Due to this dependence, we can only define derivatives at a point $x \in M$ but if we wish to define a domain for a global derivative (no dependence on a point in the manifold) our domain is what is known as the *tangent bundle*. We define this by taking the union of all tangent spaces over M. The tangent bundles will play an important role in discussing *plane fields* on our 3-manifolds; a type of subset of TM.

Definition 2.5. [7] Let M be an n-manifold. The *tangent bundle* of M, TM, is given by,

$$TM = \bigcup_{x \in M} T_xM = \{(x, v) \in M \times \mathbb{R}^n \mid v \in T_xM\}.$$

Now that we have defined what the domain and co-domain of our derivatives look like we can discuss derivatives of a function between arbitrary manifolds. Consider a pair of manifolds M and N (of dimension m, n, respectively), and some smooth map $f : M \to N$ such that for $x \in M$ we have $f(x) = y \in N$. We have claimed that we can make sense of the derivative of such a function and to do so we must make use of our charts. Suppose we have open neighbourhoods $U \subseteq M, V \subseteq N$ containing x, y respectively, with charts $\phi : U \subseteq M \to B_m \subseteq \mathbb{R}^m$ and $\psi : V \subseteq N \to B_n \subseteq \mathbb{R}^n$ for our manifolds M, N, where the B_i's are open balls. For simplicity we choose $\phi^{-1}(0) = x, \psi^{-1}(0) = y$. We have the following commutative diagram [6]:

$$\begin{tikzcd}
U \subseteq M \arrow[r, "f"] \arrow[d, "\phi"'] & V \subseteq N \arrow[d, "\psi"] \\
B_m \arrow[r, "h = \psi \circ f \circ \phi^{-1}"] & B_n
\end{tikzcd}$$

Here, the map h is simply a map of euclidean subsets and thus has derivative, dh, defined in the usual sense. The charts are homeomorphisms, whose best linear approximation are their derivatives, which are well defined. Differentiating the above commutative diagram, we get the following commutative diagram of linear mappings:

$$\begin{tikzcd}
T_xM \arrow[r, "df_x"] \arrow[d, "d\phi"'] & T_yN \arrow[d, "d\psi"] \\
\mathbb{R}^m \arrow[r, "dh_0 = d(\psi \circ f \circ \phi^{-1})_0"] & \mathbb{R}^n
\end{tikzcd}$$

As we stated above, h is a real function so its derivative at $0 \in \mathbb{R}^m$ is well defined, along with the derivatives of our charts $d\psi, d\phi$, all of which are linear mappings [6]. Thus the derivative of $f : M \to N$ is now a linear mapping of the tangents spaces of the respective manifolds, defined as the following composition:

$$df_x := d\psi^{-1} \circ dh_0 \circ d\phi. \tag{2.4}$$

The definition of both the derivative of a map and the tangent spaces are well defined no matter the choice of coordinate chart. Now that we have a notion of derivatives of functions between arbitrary manifolds, we can extend the idea of smooth functions to manifolds not living in some ambient euclidean space, which is defined in much the same way, however the partial derivatives are computed by travelling along charts associated to each manifold. We can also extend the notion of diffeomorphism to arbitrary manifolds, which gives us a notion of *equivalence* between manifolds; analogous to homeomorphisms for topological spaces or isomorphisms for groups.

Definition 2.6. [8] Let $f : M \to M$ be a smooth map, where M, N are manifolds. Then f is a *diffeomorphism* if it is bijective and f^{-1} is also smooth. In such a case, we say M and N are *diffeomorphic*, which we will denote $M \cong N$.

2.2 Regular Values and Submanifolds

Generally speaking, submanifolds are subsets of a manifold that also look locally like euclidean space. However, we also wish to use charts of our manifold, which restrict nicely to the submanifold.

Definition 2.7 (Submanifold, [9])**.** Let M be a smooth manifold with $dimM = m$ and $N \subseteq M$. Then N is a *smooth n-dimensional submanifold* of M if $\forall n \in N, \exists$ an open neighbourhood $U \subseteq M$ that contains n and a chart $\phi : U \to \mathbb{R}^m$ in M such that $\phi(N \cap U) = \phi(N) \cap \phi(U) = \mathbb{R}^n \cap \phi(U)$.

Essentially, we want charts from our manifold M that also work for points in the submanifold $N \subseteq M$. Some simple examples include, $\mathbb{R}^n$ is a submanifold of $\mathbb{R}^{n+1}$ and T^2 is a submanifold of $\mathbb{R}^3$. An important example we will use later is S^1 (or something homeomorphic to S^1 like a knot) is a submanifold of S^3. Essentially, submanifolds are topologically just subspaces, but have the extra condition they are locally diffeomorphic to euclidean space of the respective dimension. A non-example would be the Klein bottle, which is a *non-orientable* 2-manifold but is not a submanifold of $\mathbb{R}^3$. An important way of producing submanifolds from smooth maps is via pre-images of what are known as regular values.

Definition 2.8. [6] Let $f : M \to N$ be a smooth map of manifolds M, N. Then $y \in N$ is a *regular value* if the map $df_x : T_xM \to T_yN$ is surjective for all $x \in M$ with $f(x) = y$. Such an $x \in M$ is called a *regular point.*

If the above criterion does not hold for some $y \in N$, then we call it a *critical value.* Regular values will be pivotal in the Thom-Pontryagin construction, where we exploit their power in the following proposition.

Theorem 2.9 (Pre-image theorem, [6], [8])**.** *Let M, N be manifolds satisfying $dimM \geq dimN$ and $f : M \to N$ a smooth map. Then, if $y \in N$ is a regular value of f then $f^{-1}(y) \subseteq M$ is a closed submanifold of dimension $dimM - dimN$.*

Suppose we have a smooth manifold M and a smooth submanifold $M' \subseteq M$, then we have a natural smooth inclusion $i : M' \hookrightarrow M$ of which we can also form a derivative of at each $x \in M'$; $di_x : T_xM' \hookrightarrow T_xM$ which can be thought of as the inclusion of the tangent spaces. These tangent spaces, of M and $M' \subseteq M$, are in fact subspaces of one another, namely $T_xM' \subseteq T_xM$, with inclusion di_x.
We prove a helpful preparatory proposition that essentially says the restriction of the derivative (to a submanifold) is the same as the derivative of the restriction.

Proposition 2.10. *[9] Let $f : M \to N$ be a smooth map with a smooth submanifold $M' \subseteq M$. For every $x \in M'$, we have*

$$d(f|_{M'})_x = (df_x)|_{T_xM'}.$$

Proof. Let $i : M' \hookrightarrow M$ be the inclusion. Then we have,

$$d(f|_{M'})_x = d(f \circ i)_x = df_x \circ di_x = (df_x)|_{T_xM'},$$

where we are using the standard chain rule and in the final step we are using the fact di_x is the inclusion of $T_xM' \hookrightarrow T_xM$. □

Now, we recall that we can form smooth submanifolds from pre-images of regular values.

Proposition 2.11. *[9] Let $f : M \to N$ be a smooth map and $y \in N$ a regular value. Then $M' = f^{-1}(y) \subseteq M$ is a submanifold, with each $p \in M'$ satisfying,*

$$T_pM' = ker(df_p).$$

Proof. We have that $f|_{M'} = f|_{f^{-1}(y)} = y$. Taking derivatives of this at a point $p \in M'$, we get $d(f|_{M'})_p = 0$, since derivatives of constants are zero. Then by Proposition 2.10, $(df_p)|_{T_pM'} = 0$. Hence, df_p sends everything in T_pM' to zero, or in other words, $T_pM' \subseteq ker(df_p)$. To show the equality of these vector spaces, we further need only show that they are of equivalent dimension. Since $M' = f^{-1}(y)$ where y is a regular value,

$$dim(M') = dim(T_pM') = dim(M) - dim(N).$$

Then by rank-nullity theorem, we have

$$dim(ker(df_x)) = dim(T_xM) - dim(T_xN) = dim(M) - dim(N) = dim(M'),$$

where we also used that dimension of a tangent space is equivalent to the dimension of the manifold itself. Hence we have shown T_pM' is a vector subspace of $ker(df_x)$ and are of equivalent dimension, hence $T_pM' = ker(df_x)$. □

Our tangent space T_xM' and the space of vectors normal to M' in M, denoted $T_xM'^{\perp}$ form a basis for vectors in M based at $x \in M'$. That is, we can always decompose a vector in M into tangent and normal parts. From above, df_x maps T_xM' to zero and is surjective since $x = f(y)$ is a regular point, thus all of T_yN must be obtained. This leaves only the vectors in M that are normal to M' left, to be mapped isomorphically onto T_yN, hence we have an isomorphism $T_xM'^{\perp} \cong T_yN$[8]. This isomorphic mapping of the normal vectors will be pivotal in Section 5.1.

2.3 Differential Forms

Differential forms are objects we use all the time in mathematics, without ever stopping and thinking about what they are. For example, dx in the integral $\int_a^b f(x)dx$ is what is known as a *one form*, but is largely disregarded in computing integrals. Forms will need to be well understood for the definition of a contact structure so we put the definition on steady ground here.

We recall from basic group theory that for $k \in \mathbb{N}$, S_k the symmetric group on n letters which is a group whose elements are permutations of the set $\{1, \ldots, k\}$. Recall also the group homomorphism on S_k, namely the signature function $sgn : S_k \to \{\pm 1\}$, which assigns parity to the permutations.

Definition 2.12. [7] An *alternating k-form* on a real vector space V is a multilinear map $\omega : V^k \to \mathbb{R}$ satisfying

$$\omega(v_{\sigma(1)}, \ldots, v_{\omega(k)}) = sgn(\sigma)\omega(v_1, \ldots, v_k), \tag{2.5}$$

for all $v_1, \ldots, v_k \in V$ and $\sigma \in S_k$.

By multilinear, we mean that a map $T : V^k \to V^k$ is linear in each component when considering the others to be constant, i.e.

$$T(v_1, \ldots, v_i + av_i', \ldots, v_k) = T(v_1, \ldots, v_i, \ldots, v_k) + aT(v_1, \ldots, v_i', \ldots, v_k) \text{ for } a \in \mathbb{R}.$$

Alternating forms are multilinear maps that essentially only change the order of the entries in a vector and may sometimes introduce a factor of -1 depending on the parity of the choice of permutation. Alternating 0-forms are defined as real numbers and all alternating 1-forms are simply linear maps $\omega : V \to \mathbb{R}$, since nothing can alternate in one variable, hence the set of all 1-forms coincides with the dual space of the vector space. In fact, the the space of all k-forms is itself a vector space, given by,

$$\Lambda^k(V) = \{\omega : V^k \to \mathbb{R} \mid \omega \text{ is an alternating k-form}\}.$$

As stated above, $\Lambda^1(V) := V^*$, and in the case $V = \mathbb{R}^m$ considering the 1-forms $dx^i : \mathbb{R}^m \to \mathbb{R}$ which projects a vector onto its i^{th} component, i.e. $dx^i(\xi) = \xi_i$. Then $\{dx^1, \ldots, dx^m\}$ forms a basis for the dual space of $\mathbb{R}^m$. We can form new forms, of higher order, from lower order ones via what is known as the *wedge product*.

Definition 2.13 (Wedge Product, [7])**.** Let $\omega \in \Lambda^k(V^*)$ and $\tau \in \Lambda^l(V^*)$ be alternating k- and l-forms respectively. Then the *wedge product*, $\omega \wedge \tau \in \Lambda^{k+l}(V^*)$, is given by,

$$(\omega \wedge \tau)(v_1, \ldots, v_{k+l}) := \sum_{\sigma \in S_{k,l}} sgn(\sigma)\omega(v_{\sigma(1)}, \ldots, v_{\sigma(k)})\tau(v_{\sigma(k+1)}, \ldots, v_{\sigma(k+l)}). \tag{2.6}$$

Where $S_{k,l} \subseteq S_{k+l}$ is the subset of the symmetric group on $k + l$ letters, which leaves the order of the first k letters and last l letters unchanged.

This is a fairly complicated definition, a lot of which can be ignored when computing the wedge product of forms, and which simplifies tremendously for 1-forms.

Example 2.14. [7] Let $v, w \in V$ and $\alpha, \beta \in \Lambda^1(V)$ be 1-forms, then their wedge product is the 2-form $(\alpha \wedge \beta)$ is given by,

$$(\alpha \wedge \beta)(v, w) = \alpha(v)\beta(w) - \beta(v)\alpha(w). \tag{2.7}$$

Definition 2.15. Let M be a smooth manifold. A *differentiable k-form* on M is a function ω which smoothly assigns to each point $x \in M$ an alternating *k-form* $\omega(x) \in \Lambda^k(T_x^*M)$. The set of all differentiable k-forms on a manifold M is denoted $\Omega^k(M)$.

We can construct new forms in many ways, some as simple as adding; given two k-forms ω_1, ω_2 we can form $(\omega_1 + \omega_2)(x) = \omega_1(x) + \omega_2(x)$. As with alternating forms, we can also form the wedge product of differentiable forms, the only difference being we do this pointwisely. Given differentiable forms $\omega \in \Omega^k(M), \tau \in \Omega^l(M)$ then $\omega \wedge \tau \in \Omega^{k+l}(M)$ is given by,

$$(\omega \wedge \tau)(x) = \omega(x) \wedge \tau(x).$$

The final way of producing new differentiable forms is via the *exterior derivative*. Which, is the smooth operator $d : \Omega^k(M) \to \Omega^{k+1}(M)$ producing a $(k+1)$-form from a k-form. This is a generalisation of the derivative of real functions in the normal sense and hence the definitions coincide for $f \in \Omega^0(M)$. Dealing directly with the definitions when playing with differential forms is challenging and for most applications excessive. Hence, we provide the following properties below to reduce the tediousness when computing wedge products and exterior derivatives of differential forms:

Proposition 2.16. *Let $\alpha, \beta \in \Omega^k(M)$, then*

1. $d(c_1\alpha + c_2\beta) = c_1 d\alpha + c_2 d\beta$, $c_1, c_2 \in \mathbb{R}$,
2. $d(d\alpha) = 0$,
3. $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$,
4. $\alpha \wedge \beta = (-1)^{k^2} \beta \wedge \alpha$.

We will use these properties when deciding whether a one form defines a contact structure in Chapter 4, as the one form must satisfy $\alpha \wedge d\alpha > 0$. Evidently this involves the exterior derivative and the wedge product. Finally, we mention the notion of a *pullback*; an operation, which given a smooth map of manifolds $\phi : M \to N$, lets one move a one-form or smooth function from N onto M. It will feature in Section 5.1 when discussing the Thom-Pontryagin construction.

Definition 2.17. Let $\phi : M \to N$ be a smooth map of smooth manifolds and let $f : N \to \mathbb{R}$ be a smooth function. Then the *pullback* of f by ϕ, denoted $\phi^* f$, is given by

$$(\phi^* f)(x) = f(\phi(x)),\ x \in M.$$

A similar, more complicated, definition can be formulated for a general k-form. However, the idea is the same and the definition above is more than complete for our purposes in this project.

3 Open Book Decompositions

We now turn our attention to open book decompositions, which form the first class of objects in the Giroux correspondence. Throughout this section, we rely heavily on the world of knots and links, hence we begin with a word on these, before providing a series of definitions and examples.

Definition 3.1. [10] A subset K of a 3-manifold M is a *knot* if it is homeomorphic to S^1. More generally, a subset $L \subseteq M$ is a *link* if it is homeomorphic to a disjoint union $S^1_1 \uplus \cdots \uplus S^1_n$ of circles, where $\uplus$ denotes disjoint union.

Examples of nice knots and links include: the unknot, a trivial copy of S^1 and the trefoil knot, which is regarded as the simplest non-trivial knot, obtained by joining the ends of an overhand knot. A relevant example to the project are the *Hopf links*, which can be achieved by attaching two unknots such that they cross exactly once. One can do exactly this with three unknots to achieve what are called the *Borromean rings*. This idea of links crossing, introduces an invariant of knots known as the linking number. This characterises the number of times two knots wind around one another, takes an integer value and can be positive or negative depending on the relative orientations of the knots. This can be seen in Figure 2, displaying the positive and negative Hopf links, with H^+ having linking number $+1$ and H^- linking number -1.

3.1 Definitions and Examples

Definition 3.2. [11] An *open book decomposition* (OBD) of a closed, oriented 3-manifold, M, is a pair (B, π) such that:

1) B is a link in M and,
2) π is a map (fibration) $\pi : M \setminus B \to S^1$ where $\pi^{-1}(\theta)$ is the interior of a compact surface Σ_θ, whose boundary is $\partial\Sigma_\theta = B$ for each $\theta \in S^1$.

The surfaces Σ_θ are called the pages of the OBD, whilst B is the binding. The image one should hold in mind, and as the name *open book* suggests, is a book being opened and the front and back cover being glued to each other. Then the spine of the book acts as our binding, whilst the pages extrude out, one for each $\theta \in S^1$. We give two examples of such OBDs for the 3-sphere, $M = S^3$, which are realised using the unknot and the Hopf links.

Example 3.3. [11] We provide examples of open books for our manifold $M = S^3$. It is common practice to regard $M = S^3$ as the unit sphere in $\mathbb{C}^2$, which has coordinates $(z_1, z_2) = (r_1 e^{i\theta_1}, r_2 e^{i\theta_2})$, by defining $S^3 = \{(z_1, z_2) \in \mathbb{C}^2 \mid |z_1|^2 + |z_2|^2 = 1\}$.

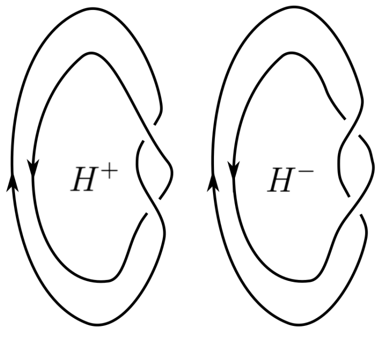

Figure 2: The positive and negative Hopf links.

1. Consider $U = \{z_1 = 0\} \cap S^3 \subseteq S^3$, which is just a unit S^1 in S^3 since $U = \{(0, e^{i\theta_2})\}$, in fact U is just an unknot in S^3. This has fibration,
$$\pi_U : S^3 \setminus U \to S^1 \text{ given by } (z_1, z_2) \mapsto \frac{z_1}{|z_1|}, \tag{3.1}$$
which in polar coordinates is $\pi_U(z_1, z_2) = \theta_1$. Thus, (U, π_U) is an open book for S^3, with a simple unknotted binding. Recall that, the pages have boundary B, hence the only shape with S^1 boundary is the 2-disc. Hence the pages resemble open discs and can be parametrised by,
$$\pi_U^{-1}(\theta) = \{(z_1, z_2) \in S^3 \mid |z_2| < 1, z_1 = \sqrt{1 - |z_2|^2} e^{i\theta}\}. \tag{3.2}$$

2. A slightly more interesting example is provided by choosing our binding to be the positive or negative Hopf link, as seen in Figure 2. We define the positive and negative Hopf links in S^3 by
$$H^+ = \{(z_1, z_2) \in S^3 \mid z_1 z_2 = 0\} \text{ and } H^- = \{(z_1, z_2) \in S^3 \mid z_1 \overline{z}_2 = 0\}, \tag{3.3}$$
with fibrations,
$$\pi_\pm : S^3 \setminus H^\pm \to S^1 : \pi_+(z_1, z_2) = \frac{z_1 z_2}{|z_1 z_2|} \text{ and } \pi_-(z_1, z_2) = \frac{z_1 \overline{z}_2}{|z_1 \overline{z}_2|}. \tag{3.4}$$
If one changes to polar coordinates, for H^+ we get,
$$\pi_+(z_1, z_2) = \frac{z_1 z_2}{|z_1 z_2|} = \frac{r_1 r_2 e^{i(\theta_1 + \theta_2)}}{|r_1 r_2|} = e^{i(\theta_1 + \theta_2)},$$
in exponential form, which always has has magnitude 1 so has argument $\theta_1 + \theta_2$. A similar calculation follows for H^-, yielding $\pi_\pm(r_1 e^{i\theta_1}, r_2 e^{i\theta_2}) = \theta_1 \pm \theta_2$ in polar coordinates. For (H^+, π_+), we can parametrise the pages by
$$\pi_+^{-1}(\phi) = (z_1 = \sqrt{t} e^{i\theta}, z_2 = \sqrt{1-t} e^{i(\phi - \theta)}) \text{ for } t \in (0,1), \theta \in S^1.$$
As expected, given our binding is H^+, this shows our pages to be annuli. A similar calculation can be carried out for (H^-, π_-). This gives two *different* OBDs for S^3, where the binding is the positive or negative Hopf link and we have annuli as pages.

It will also prove useful to consider open books abstractly by a pair (Σ, ϕ) from which we can reconstruct our manifold. This is a slightly weaker definition of an OBD since we can only talk about abstract open books up to diffeomorphism in M, whereas with the standard definition, one can talk of open books up to isotopy. In practice, one interchanges between the notions to suit the type of problem without much care.

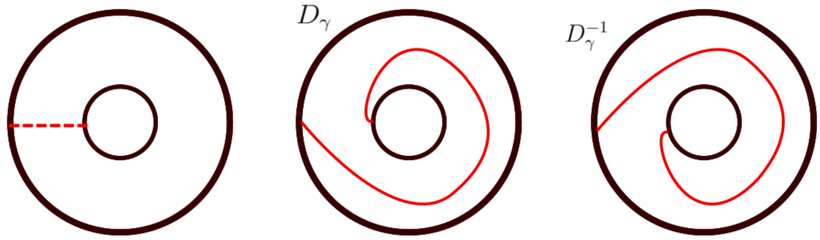

Figure 3: The right handed and left handed Dehn twist on an annulus.

Definition 3.4. [12] An *abstract open book* for a 3-manifold, M, is a pair (Σ, ϕ) such that:

1) Σ is a compact surface with boundary
2) $\phi : \Sigma \to \Sigma$ is a diffeomorphism, called the monodromy, which restricts to the identity on $\partial\Sigma$.

With the abstract data, we can reconstruct a manifold as follows. We consider $\Sigma \times [0,1]$ and first glue points in the manifold at either end of the interval, subject to the monodromy. That is, we relate, $(\phi(x), 0) \sim (x, 1)$, for all $x \in \Sigma$. Then we identify points of the boundary of our surface via $(y, 0) \sim (y, t), \forall t \in [0,1], y \in \partial\Sigma$. This construction gives us our 3-manifold:

$$M = \frac{\Sigma \times [0,1]}{\begin{array}{c}(\phi(x), 0) \sim (x, 1)\ \forall x \in \Sigma, \\ (y, 0) \sim (y, t)\ \forall y \in \partial\Sigma, t \in [0,1]\end{array}}. \tag{3.5}$$

If we consider $q : \Sigma \times [0,1] \to M$ to be our quotient map, then we can regard $q(\partial\Sigma)$ as the binding and $q(\Sigma \times \{t\})$ as a page of our open book, one for each $t \in [0,1]$ [12].We provide some examples of such a construction:

Example 3.5. Consider the abstract pair $(\Sigma, \phi) = (D^2, id)$. Then when identifying points $(x, 0) \sim (x, 1)$ in $[0,1] \times D^2$ we produce the solid torus, $S^1 \times D^2$. From here we identify points $(y, 0) \sim (y, t)$ for all $t \in [0,1]$ and $y \in \partial D^2$. This process caps off the torus to produce S^3.
Consider the abstract pair of A the closed annulus with identity monodromy. We cross A with $[0,1]$ and then follow the same process to give us our 3-manifold, this time $S^1 \times S^2$.

In the two examples above we notice that the monodromy is simply the identity, however this need not be the case. For the next example, we consider a non-trivial monodromy where in both cases we use a very important function, namely the *Dehn twist*. Not only will this provide interesting examples of OBDs, but it will also later on help establish the stabilisation of open books in Section 3.3.

Dehn Twists [11] Consider a surface Σ and a curve, γ, properly embedded in Σ and $N = \gamma \times (0,1)$, an open neighbourhood of γ. We define the right handed Dehn twist, D_γ, to be the diffeomorphism that is the identity on $\Sigma \setminus N$ and is given by $(\theta, t) \mapsto (\theta + 2\pi t, t)$ on N. The left handed Dehn twist, denoted D_γ^{-1}, is again the identity on $\Sigma \setminus N$ whilst on N is given by, $(\theta, t) \mapsto (\theta - 2\pi t, t)$. The left and right handed Dehn twists on a closed annulus can be seen in Figure 3.

Example 3.6. Consider an annulus, A, with the right handed Dehn twist as our monodromy, i.e $(\Sigma, \phi) = (A, D_\gamma)$. $A \times [0,1]$ is again a cylindrical annulus, however now when identifying $(x, 0) \sim (\phi(x), 1)$ for $x \in A$ we are applying a right handed Dehn twist for points in the interior. We then identify points on the boundary, both the outer and inner boundaries of A, for all $t \in [0,1]$. This results in constructing S^3. We could have also done this with a left handed Dehn twist, D_γ^{-1}, from which we would've achieved another open book decomposition for S^3. However, the OBD achieved from D_γ and D_γ^{-1} are fundamentally different; they are not diffeomorphic.

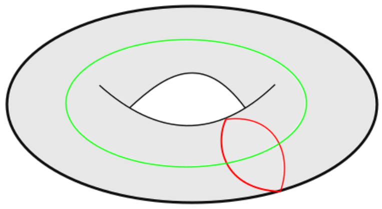

Figure 4: The solid torus with a choice of meridian (red) and longitude (green) curves.

3.2 Existence of Open Books on 3-Manifolds

Theorem 3.7 (Alexander, [2], [11], [13]). *Every closed, oriented, 3-manifold M has an open book decomposition.*

Above we have stated a powerful theorem on the existence of OBDs on 3-manifolds, the proof of which relies on the machinery of *Dehn surgery* [10]. This elegant technique is capable of producing all 3-manifolds, via surgery on a link. The surgery process can be broken down into two parts, namely Dehn *drilling* and Dehn *filling*. Suppose we have a 3-manifold M and a link $L \subseteq M$ in the interior of M. We take an open tubular neighbourhood of our link, $N(L) = L \times int(D^2)$, and remove it from M, giving the manifold *drilled* along L with boundary, $\partial M \cong L \times S^1$. This divides the manifold into two; the tubular neighbourhood of our link $N(L)$, which we identify with a solid torus, $S^1 \times D^2$, and $M \setminus N(L)$, the knot exterior.
On a torus, there are two pivotal closed curves, namely the meridian, m, and the longitude, l. Any simply closed curve on the torus can be decomposed into integer multiples of m and l [10].

Definition 3.8. [10] The *meridian*, m, and *longitude*, l, curves are the generators for the fundamental group of the torus, that is, $\langle m, l\rangle = \pi_1(T^2)$. Alternatively, for a solid torus, $S^1 \times D^2$, we identify the meridian with the curve on $\partial(S^1 \times D^2)$ which bounds a disc in $S^1 \times D^2$ and the longitude to be any curve on $\partial(S^1 \times D^2)$ which intersects m once.

This definition leaves a lot of room for choice, but typically one identifies the meridian with $m = \{0\} \times \partial D^2 \subseteq S^1 \times \partial D^2$, before choosing a longitude that intersects m once. A choice of such a meridian and longitude can be seen in Figure 4.
The Dehn filling involves gluing $N(L)$ back into M, by gluing the meridian curve of $N(L)$ to a simply closed curve, γ, on $S^1 \times D^2$. Gluing in $N(L)$ exactly as it was (gluing the meridian of $N(L)$ to where it was before) leads to no change of M. We can have more exotic gluings, and hence changes to M, from our choice of γ. As stated, we can write γ as having gone a times round the meridian plus b times round the longitude. Informally, one can think of this as a type of linear combination, that is we can write $\gamma = a \cdot m + b \cdot l$. Every possible gluing is determined by these integers $a, b \in \mathbb{Z}$, whose ratio $\frac{a}{b} \in \mathbb{Q} \cup \{\infty\}$ is called the surgery coefficient (we allow infinity as we may have $b = 0$). The power of this Dehn surgery is captured in the following fundamental theorem.

Theorem 3.9 (The fundamental theorem of surgery). *[13] Every closed, oriented,* 3*-manifold can be obtained by surgery on a link in S^3. A representation of this surgery can always be found with a* ± 1 *surgery coefficient and with unknots as the individual disjoint components of the link.*

A well known result of this fundamental theorem is that it is possible to construct an unknot which traverses each component of the link precisely once. We regard this unknot as U. In such a case, we say that the link, L, is *braided* about U [11]. The idea is analogous to threading a rope through each unknotted component of the link.
Having introduced links and the theorem of surgery, we can now prove Theorem 3.7. We will use the open book (U, π_U) for S^3, as seen in Example 3.3.

Proof of Theorem 3.7. We perform surgery along a link in S^3 which have tubular neighbourhoods we identify with disjoint solid tori $V_1, \ldots, V_r$. From Theorem 3.9, we can represent this as a $+1$ surgery to produce a manifold M which contains the disjoint solid tori $V_1', \ldots, V_r'$. From this surgery, we obtain a homeomorphism:

$$h : S^3 \setminus (\mathring{V}_1 \uplus \cdots \uplus \mathring{V}_r) \to M \setminus (\mathring{V}_1' \uplus \cdots \uplus \mathring{V}_r'),$$

which maps meridians of V_i onto longitudes of V_i', where $\mathring{V}$ denotes the interior of the tubular neighbourhood of the knot. We may safely assume, each V_i wraps around some unknot $U \subseteq S^3$ in a neat way (braided) such that each V_i intersects the pages of the open book (U, π_U) along meridional discs of V_i. Then the binding of the OBD for M will consist of $h(U) \uplus c_i$, where the c_i are the central curves [1] of V_i'. Then the new pages have boundary $h(U)$ (which is still homeomorphic to U) and c_i, hence the pages intersect ∂V_i in longitudinal curves. We can extend the fibration π_U onto a fibration of $M \setminus (c_1 \uplus \cdots \uplus c_r \uplus)$ given by:

$$\pi_U \circ h^{-1} : M \setminus (\mathring{V}_1' \uplus \cdots \uplus \mathring{V}_r' \uplus h(U)) \to S^1.$$

Hence, we have an open book decomposition for M with binding $h(U) \uplus c_i$ and fibration $\pi_U \circ h^{-1}$. $\square$

3.3 Stabilisations of Open Books

In the Giroux correspondence, we fix the uniqueness of open books *supporting* a contact structure, by talking of OBDs up to positive stabilisation, that is, if (Σ', ϕ') is an open book obtained from (Σ, ϕ) via a positive stabilisation, then they both correspond to the same contact structure in the Giroux correspondence. Later, we will also think about *negative* stabilisations and their affect on the associated contact structures.
To conduct a stabilisation of an open book (Σ, ϕ) we first add what is known as a 1-handle (h_1) to each page, to produce a new page $\Sigma' = \Sigma \cup h_1$. This process produces a new page with a U-shaped handle; analogous to the handle on a kettlebell one finds in a gym. We then find a curve which traverses the handle exactly once and compose our monodromy with a (left or right) Dehn twist along this curve. This is all summarised in the following definition:

Definition 3.10. [14] A positive (negative) *stabilisation* of an open book (Σ, ϕ) is the open book with,

1. new pages $\Sigma' = \Sigma \cup h_1$
2. new monodromy $\phi' = \phi \circ \tau_c$, where τ_c is the right (left) handed Dehn twist along a curve c which traverses the 1-handle, h_1, exactly once.

The resulting stabilisation we denote by $S_\pm(\Sigma, \phi)$. Stabilisation is actually a special case of another operation, we call the *Murasugi sum*, which lets one construct new open books.

Definition 3.11 (Murasugi sum [11])**.** Given two open books $(\Sigma_0, \phi_0), (\Sigma_1, \phi_1)$, properly embedded curves c_0, c_1 with rectangular neighbourhoods $R_0 = c_0 \times [-1, 1], R_1 = c_1 \times [-1, 1]$. Then the *Murasugi sum* of the open books is $(\Sigma_0, \phi_0) * (\Sigma_1, \phi_1)$ with pages,

$$\Sigma = \Sigma_0 \cup_{R_0 = R_1} \Sigma_1,$$

where $\cup_{R_0=R_1}$ means gluing by sending $\partial c_1 \times [-1, 1] \mapsto c_2 \times \{-1, 1\}$ and $c_1 \times \{-1, 1\} \mapsto \partial c_2 \times [-1, 1]$, and monodromy map $\phi^* = \phi_0 \circ \phi_1$.

[1] Where the central curves refer to the core of the tubular neighbourhood, in fact exactly the knot of whom we are taking a tubular neighbourhood of. More precisely, it is the curve $S^1 \times \{\mathbf{0}\}$ in $S^1 \times D^2$.

Using Murasugi sums on open books we have the following well-known theorem.

Theorem 3.12. *[11] Let* $(\Sigma_0, \phi_0), (\Sigma_1, \phi_1)$ *be open books, then*

$$M_{(\Sigma_0,\phi_0)} \# M_{(\Sigma_1,\phi_1)} \cong M_{(\Sigma_0,\phi_0)*(\Sigma_1,\phi_1)}.$$

In summary, the manifold associated to the Murasugi sum of the open books $(\Sigma_0, \phi_0), (\Sigma_1, \phi_1)$ is diffeomorphic to the connected sum of the manifolds associated to each open book respectively. The connected sum is defined in the usual sense; by punching out of a 3-ball in each 3-manifold and gluing along the respective boundary spheres.
Having now seen the notions of stabilising open books and producing new ones via Murasugi sums, we have the machinery to show a somewhat surprising fact that stabilising an open book is equivalent to undertaking a Murasugi sum with a Hopf link and its associated fibration (this is sometimes called *plumbing with a Hopf band)*, as seen in Example 3.3.

Proposition 3.13. *[11] Let* (Σ, ϕ) *be an open book, then*

$$S_{\pm}(\Sigma, \phi) = (\Sigma, \phi) * (H^{+}, \pi_{+}).$$

Proof. This is simply an application of the definitions. Consider curves $c \in \Sigma$, $c_H \in H^{\pm}$, where c_H is the core curve of the Hopf band. We Murasugi sum along rectangular neighbourhoods of these curves to produce a new page with a one handle with a twist, from the Hopf link. Any curve that then traverses this handle is equivalent to one composed with a Dehn twist. An excellent motivating illustration of this can be found in [15]. □

Given Proposition 3.13 and Theorem 3.12, we can now show that stabilising an open book does in fact not change the manifold associated to the open book itself.

Theorem 3.14. *[11] Let* (Σ, ϕ) *be an open book, then*

$$M_{S_{\pm}(\Sigma,\phi)} \cong M_{(\Sigma,\phi)}.$$

Proof. By Proposition 3.13 and Theorem 3.12, we have that $M_{S_{\pm}(\Sigma,\phi)} \cong M_{(\Sigma,\phi)*(H^{\pm},\pi_{\pm})} \implies M_{S_{\pm}(\Sigma,\phi)} \cong M_{(\Sigma,\phi)} \# M_{(H^{\pm},\pi_{\pm})}$. By Example 3.3 we have that the manifold associated to the open book $(H^{\pm}, \pi_{\pm})$ is S^3. Hence, $M_{S_{\pm}(\Sigma,\phi)} \cong M_{(\Sigma,\phi)} \# S^3$ but taking the connected sum of any 3-manifold with S^3 leaves the manifold unchanged. Thus, $M_{S_{\pm}(\Sigma,\phi)} \cong M_{(\Sigma,\phi)}$. □

4 Contact Structures

We now turn our attention to the second class of objects in the Giroux correspondence, namely contact structures. Contact structures are a sub-class of the collection of 2-*plane fields* on a 3-manifold, which have the following definition.

Definition 4.1. [16] Let M be a 3-manifold, then a 2-*plane field*, ξ, on M is a 2-dimensional sub-bundle of the tangent bundle, TM, such that for each point $p \in M$, we have a two dimensional subspace $\xi(p) \subseteq T_pM$.

All plane fields can be locally described by the kernel of a 1-form, $\alpha \in \Omega^1(M)$, which is useful when discussing the integrability of such plane fields. In fact, a plane field given by $\xi = ker(\alpha)$ is integrable exactly when the following Frobenius integrability condition is satisfied,

$$\alpha \wedge d\alpha \equiv 0. \tag{4.1}$$

Contact structures are in some sense the converses of such integrable plane fields, as they are the *maximally non-integrable* plane fields on a manifold M.

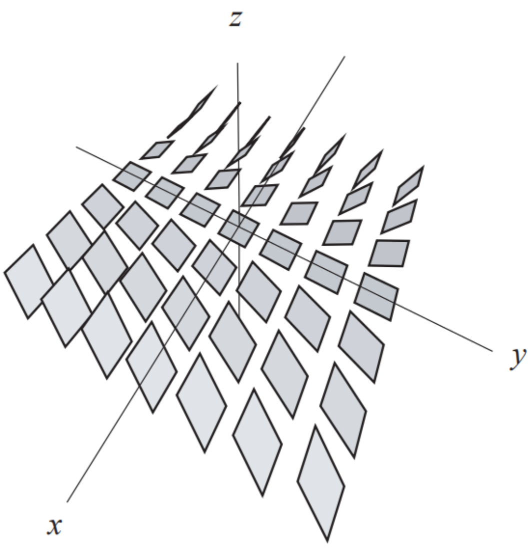

Figure 5: The standard contact structure, ξ_1, on $\mathbb{R}^3$ [16].

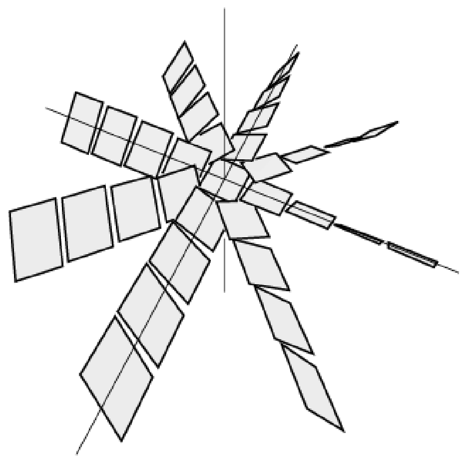

Figure 6: The contact structure ξ_2 on $\mathbb{R}^3$ [11].

Definition 4.2. A *contact structure* on a 3-manifold M is a maximally non-integrable plane field ξ, $\xi = ker\ \alpha$ with the 1-form α satisfying,

$$\alpha \wedge d\alpha > 0. \tag{4.2}$$

The pair (M, ξ) is called a contact manifold and such a 1-form, α, is called a contact form.

Note that we could have changed 4.2 to $\alpha \wedge d\alpha \not\equiv 0$ and had a similarly defined maximally non-integrable plane field. What we have defined above is, in fact, a *positive* contact structure; however, the difference is negligible for our construction. We provide some examples of contact structures in $\mathbb{R}^3$.

Example 4.3. [11] We begin with what is known as the standard contact structure in $\mathbb{R}^3$. Working in Cartesian coordinates (x, y, z), define:

$$\alpha_1 = dz + xdy. \tag{4.3}$$

Then $d\alpha_1 = dx \wedge dy$ and so $\alpha_1 \wedge d\alpha_1 = dz \wedge dx \wedge dy$ which is certainly non-zero as this is the standard volume form in $\mathbb{R}^3$. Hence, $\alpha_1 \wedge d\alpha_1 > 0$ and α_1 is a contact form. We see that, at a point $(x, y, z) \in \mathbb{R}^3$,

$$\xi_1 = ker(\alpha_1) = span\left\{\frac{\partial}{\partial x}, x\frac{\partial}{\partial z} - \frac{\partial}{\partial y}\right\}.$$

Notice, that for $x = 0$, ξ is spanned by $\{\frac{\partial}{\partial x}, -\frac{\partial}{\partial y}\}$, so our planes are horizontal in the y-z plane. As we increase in x, starting from 0, the planes slowly twist in a clockwise direction, but don't even make a 90 degree rotation as $x \to \infty$. The described behaviour can be seen in Figure 5, which shows what is happening close to the x-y plane. The behaviour can be extrapolated above and below, perpendicular to the y-z plane.

Example 4.4. [16] Considering $\mathbb{R}^3$ with cylindrical coordinates (r, θ, z) and the contact form,

$$\alpha_2 = dz + r^2 d\theta. \tag{4.4}$$

At a point (r, θ, z), ξ_2 is spanned by $\{\frac{\partial}{\partial r}, r^2\frac{\partial}{\partial z} - \frac{\partial}{\partial \theta}\}$. Once again, along $r = 0$, ξ_2 is horizontal and again twists clockwise as r increases. Such a contact structure can be seen in Figure 6. Those with a keen eye for detail will notice that ξ_2 looks like a slice of ξ_1 along a fixed x value, rotated around the z-axis. This is because there are, in fact, the same contact structure, up to isotopy.

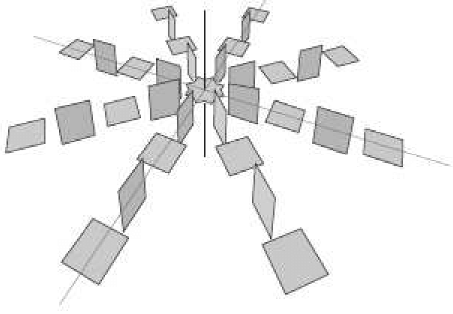

Figure 7: The standard overtwisted contact structure on $\mathbb{R}^3$ [16].

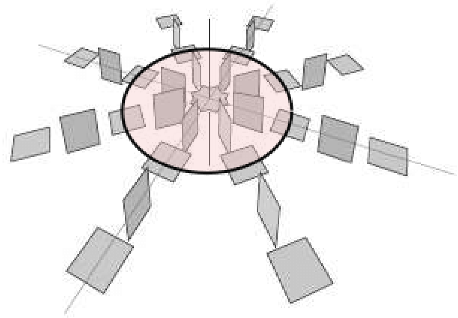

Figure 8: The standard overtwisted contact structure in $\mathbb{R}^3$ with the disc of radius $r = \pi$; the standard overtwisted disc.

Example 4.5. [11] We provide another example of a contact structure, again on $\mathbb{R}^3$ in cylindrical coordinates (r, θ, z), which is fundamentally different to Examples 4.3 and 4.4. We define the one form,

$$\alpha_{ot} = cosr\, dz + r\, sinr\, d\theta. \tag{4.5}$$

One can compute the wedge, $\alpha_{ot} \wedge d\alpha_{ot} = (1 + \frac{sinr}{r} cosr) rdr \wedge d\theta \wedge dz$, where at $r = 0$ we let $\frac{sinr}{r} = 1$, which is always positive so α_{ot} is a contact form. Thus the contact structure is,

$$\xi_{ot} = ker(\alpha_{ot}) = span\left\{\frac{\partial}{\partial r}, r\, sinr\, \frac{\partial}{\partial z} - cosr \frac{\partial}{\partial \theta}\right\}.$$

Notice, that much like ξ_1, ξ_{ot} is horizontal along the z-axis (at $r = 0$) and both contact structures also rotate as r increases. However, ξ_{ot} makes infinitely many rotations as $r \to \infty$, and at integer multiples of π, we get the contact structure tangent to the x-y plane, as seen in Figure 5.

Above, I have claimed that even though each of ξ_1, ξ_2 and ξ_{ot} are all defined differently, that ξ_1 and ξ_2 are the same contact structure (up to isotopy), while ξ_{ot} is fundamentally different from both. The notion through which we decide equivalence is a morphism called a *contactomorphism*, which is the analogue of diffeomorphisms for manifolds or isomorphism for groups.

Definition 4.6. Let ξ_0, ξ_1 be contact structures on a manifold M. A diffeomorphism $f : M \to M$ that preserves the contact structure is a *contactomorphism*. That is, f satisfies

$$f_*(\xi_0) = \xi_1.$$

We now arrive at a surprising fact relating to the local structure of contact structures, namely all contact structures, even those which are *not* contactomorphic, are locally identical. This is captured in the following theorem, namely *Darboux's theorem*.

Theorem 4.7 (Darboux)**.** *Let (M, ξ) be any contact 3-manifold and $a \in M$. Then there exists a neighbourhood N of $a \in M$, and a neighbourhood $N' \subseteq \mathbb{R}^3$ around $(0, 0, 0)$ and a contactomorphism,*

$$f : (M, \xi|_N) \to (\mathbb{R}^3, \xi_1|_{N'}).$$

Where ξ_1 is the standard contact structure on $\mathbb{R}^3$; as defined in Example 4.3.

This is a somewhat surprising fact, but is related to the fact that local neighbourhoods of points in contact manifolds just look like flat discs. Hence ξ_1 and ξ_{ot} look identical on a local scale, although they are not contactomorphic. Thus, any interesting properties must relate to the global topology of the contact manifold.

4.1 The Overtwisted-Tight Dichotomy

Contact structures can be divided into two classes, namely *overtwisted* and *tight*, relating to properties of a global nature. The contact structures ξ_1 and ξ_{ot} fall into opposing classes, with ξ_1 being tight and ξ_{ot} being overtwisted. We saw in Example 4.5 that the contact structure, ξ_{ot}, rotates and does a half-flip every integer multiple of π, so ξ_{ot} is tangent to the horizontal plane here. Let us define the disc $\Delta = \{r \leq \pi, z = 0\} \subseteq \mathbb{R}^3$, then $\partial\Delta$ is tangent to ξ_{ot}, as seen in Figure 8. Such a disc Δ, those that are tangent to the contact structure along their boundary, are precisely the overtwisted discs.

Definition 4.8. [17] Let (M, ξ) be a contact manifold. A disc Δ embedded in the contact manifold is an *overtwisted disc* if $T_p\Delta = \xi_p$ for all $p \in \partial\Delta$.

Whether a contact structure falls into the overtwisted or tight class depends precisely on whether such a disc Δ can be embedded into the contact manifold.

Definition 4.9. Let (M, ξ) be a contact manifold. The contact structure, ξ, is *overtwisted* if it contains an overtwisted disc. The contact structure is *tight* if it is not overtwisted.

4.2 The Lutz Twist

Given any contact structure, we can in fact transform it into one that is overtwisted by performing what is known as a *Lutz twist* [17]. Let $K \subseteq (M, \xi)$ be a knot embedded in the contact manifold such that it is transverse to ξ and has the same orientation as the contact structure. Such a knot is called a *positive transverse knot*. We can then form a tubular neighbourhood of K and identify K with

$$S^1 \times \{\mathbf{0}\} \subseteq S^1 \times D^2,$$

with coordinates $\theta \in S^1$ and $(r, \phi) \in D^2$. Then the contact form on the neighbourhood of K, $S^1 \times D^2$, is $\alpha = d\theta + r^2 d\phi$, as similarly seen in Example 4.4, and ∂_θ defines the positive orientation of K. We construct another contact form on $S^1 \times D^2$, defined as

$$\alpha' = f_1(r)d\theta + f_2(r)d\phi. \tag{4.6}$$

Where $(f_1(r), f_2(r)) = (1, r^2)$ near $r = 1$, $(f_1(r), f_2(r)) = (-1, -r^2)$ near $r = 0$ and $(f_1(r), f_2(r))$ is never parallel to $(f_1'(r), f_2'(r))$, i.e. $f_1 f_2' - f_2 f_1' > 0$. A parametric graph of (f_1, f_2) with respect to r would start at (-1, 0) and curve round anti-clockwise before moving upwards at some $r = r_0$, given by $f_2(r_0) = 0$. This can be seen in Figure 9.

Definition 4.10. [18] Let (M, ξ) be a contact manifold and $K \subseteq (M, \xi)$ a positively transverse knot. Then, replacing $\xi = ker(\alpha)$ with $\xi' = ker(\alpha')$ on a neighbourhood of K is called performing a Lutz twist along K.

Corollary 4.11. *[18] Performing a Lutz twist on a contact structure results in an overtwisted contact structure.*

Proof. Let $r_0 \in (0, 1)$ be the point where $f_2(r_0) = 0$. Then at a point in $S^1 \times D^2$, $p = (\theta_0, (r_0, \phi))$, we have that the contact form reduces to $\alpha' = f_1(r)d\theta$ and hence $\xi = ker(\alpha')$ is spanned by $\{\partial_\phi, \partial_r\}$ which produces the r-ϕ plane and is thus tangent to the disc in the $\theta = \theta_0$ plane given by, $D_0 = \{(\theta_0, (r, \phi)) \mid r \leq r_0, \phi \in S^1\}$. Since this embedded disc is tangent to ξ at the point p it is an overtwisted disc. □

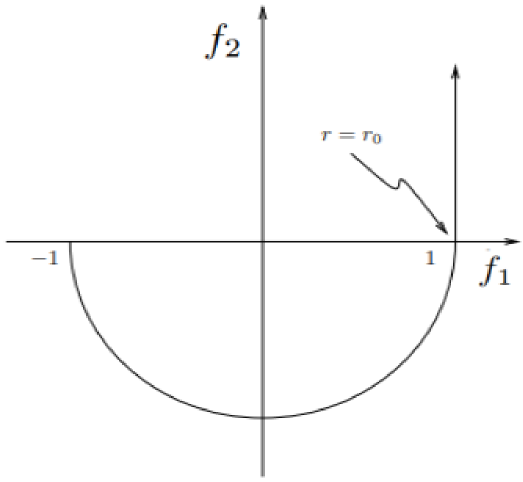

Figure 9: A parametric description of the Lutz twist.

5 2-Plane Fields on 3-Manifolds

In this chapter, we take a step back and discuss the more general notion of 2-plane fields, 2-dimensional sub-bundles of the tangent bundle of M, of which contact structures form a sub-class. The motivation for this is that 2-plane fields are far easier to work with and their classification into homotopy classes is well understood. We begin by stating a well-known and powerful theorem about tangent bundles of 3-manifolds.

Theorem 5.1. *[17] Every closed, orientable, 3-manifold has a trivial tangent bundle.*

Proof. A proof is beyond the scope of this project. However, three "simple" proofs can be found in [19]. □

A manifold that admits a trivial tangent bundle is known as *parrallelisable*. In such a case, we have an isomorphism $TM \cong M \times \mathbb{R}^3$. Equivalently, a parrallelisable n-manifold has a collection of smooth vector fields $\{V_1, \dots, V_n\}$, such that at each point $p \in M$, $\{V_1(p), \dots, V_n(p)\}$ forms a basis for the tangent space T_pM. In the case M is parrallelisable, we can also trivialise a certain sub-bundle of the tangent bundle, namely the *unit* tangent bundle, STM, which is simply the restriction of TM to just vectors of unit length in the tangent space,

$$STM = \{(x, v) \mid x \in M, v \in T_xM, ||v|| = 1\}.$$

Since T_xM forms a vector space of dimension 3, the set $\{v \in T_xM \mid ||v|| = 1\}$ acts as the two-sphere inside T_xM, $S^2 \subseteq T_xM$. By the parrallelisability we can then achieve an isomorphism $STM \simeq M \times S^2$. Since functions are really just ordered pairs; there is a corresponding map $f : M \to S^2$, giving a one-to-one correspondence of homotopy classes of

$$\{\text{2-plane fields on M}\} \longleftrightarrow \{\text{Maps } f : M \to S^2\}.$$

The map f_ξ associated to the 2-plane field ξ is a Gauss map, which maps each $p \in (M, \xi)$ to a unit vector orthogonal to M at p.

5.1 The Thom-Pontryagin Construction

The following construction will then put homotopy classes of maps $f : M \to S^2$ in one-to-one correspondence with framed cobordism classes of links in our manifold M. The maps $f : M \to S^2$, we understand fairly well, however we need a few definitions about links, which are submanifolds of M, to understand the likely foreign terms *framed* and *cobordism.*

Definition 5.2. [17] Let M be a 3-manifold and $L_0, L_1 \subset M$ be links in M. Then L_0 is *cobordant* to L_1 if $\exists$ a surface $S \subseteq [0,1] \times M$, such that

$$\partial S = L_0 \uplus L_1 \subseteq \{0\} \times M \uplus \{1\} \times M.$$

This gives an equivalence relation on submanifolds (in our case links) of the same dimension. The surface S is referred to as a *cobordism.* If two links are cobordant then they live in the same cobordism class.
The idea of *framing* is used widely in differential geometry and amounts to constructing a basis. In the context of links, a framing can be thought of as giving the link a thickness, such that we no longer think of the link as a finite number of copies of S^1, but now made up of solid tori $S^1 \times D^2$. We do this by providing a basis to vectors normal to the link.

Definition 5.3. [8] A *framing* of a link $L \subset M$ is a smooth function $\mathfrak{b}$ such that,

$$\mathfrak{b}(x) = (b^1(x), b^2(x)),$$

is a basis for the space of normal vectors to L at x, $T_xL^{\perp}$.

Given a link, L, and a framing, $\mathfrak{b}$, we call the pair $(L, \mathfrak{b})$ a framed link. The correspondence between maps and links is constructed as follows.
Consider some map $f : M \to S^2$ and a regular value $y \in S^2$. By Theorem 2.9, $f^{-1}(y)$ is a submanifold of M with dimension $dim(M) - dim(S^2) = 1$. These manifolds must be closed, hence $f^{-1}(y)$ is a link in M. Let us denote the corresponding link, $L_f = f^{-1}(y)$. This map also induces a framing of the link, L_f, by pulling back a basis of T_yS^2. Suppose $\mathfrak{b}(x) = (b^1(x), b^2(x))$ is a basis for T_yS^2, with $y \in S^2$ a regular value. Then consider for each $x \in f^{-1}(y)$ the linear map,

$$df_x : T_xM \to T_yS^2.$$

By Proposition 2.11, each point in the subspace $T_xf^{-1}(y)$ is mapped to zero in T_yS^2, which leaves the orthogonal complement, $T_xf^{-1}(y)^{\perp}$, to be mapped isomorphically onto T_yS^2. Hence for each $v^i \in \mathfrak{b}$, the basis for T_yS^2, there is a unique vector $w^i \in T_xf^{-1}(y)^{\perp}$ that maps to v^i under df_x. Thus, we have a basis for the normal bundle of our link $L_f = f^{-1}(y)$, which we denote $f^*\mathfrak{b} = (w^1(x), w^2(x))$, and hence a framed link $(L_f, f^*\mathfrak{b})$.
One may be thinking that since the construction is dependent on the choice of regular value, $y \in S^2$ and basis $\mathfrak{b}$, we should get many different framed links corresponding to a single choice of map. This is most certainly true, however, all the framed links corresponding to a single map f belong to the same framed cobordism class, regardless of our choices of y and $\mathfrak{b}$.

Proposition 5.4. *[8] Let $y, y' \in S^2$ be regular values of the map $f : M \to S^2$ and $\mathfrak{b}, \mathfrak{b}'$ bases for T_yS^2 and $T_{y'}S^2$ respectively. Then $(f^{-1}(y), f^*\mathfrak{b})$ is framed cobordant to $(f^{-1}(y'), f^*\mathfrak{b}')$.*

To show how we go from a framed link to a map $f : M \to S^2$ we rely heavily on the following important theorem.

Theorem 5.5 (Product neighbourhood theorem, [8])**.** *Let $L \subseteq M$ be a link, then some neighbourhood, N_L, of the link is diffeomorphic to $L \times \mathbb{R}^2$.*

Using the product neighbourhood theorem, we choose a product representation of a neighbourhood of our link, $g : L \times \mathbb{R}^2 \to N_L$ and consider the projection map $\pi : N_L \to \mathbb{R}^2$ given by $\pi(g(x, y)) = y$. We see 0 is a regular value of this map and $\pi^{-1}(0)$ is the link itself with the given framing. Then choose a map $\phi : \mathbb{R}^2 \to S^2$ such that $\phi(x) = p$ for $||x|| \geq 1$, where p is some base point on the sphere, and ϕ sends everything else, i.e the open unit disc, diffeomorphically onto $S^2 \setminus \{p\}$, which is of course homeomorphic to a disc. Then we define the map $f_L : M \to S^2$ by

$$f_L(x) = \begin{cases} \phi(\pi(x)) & x \in N_L \\ p & x \notin N_L, \end{cases} \tag{5.1}$$

which is the corresponding map in the one-to-one correspondence [8]. In simple terms, the corresponding map to a framed link is the one which sends a tubular neighbourhood of the link to the sphere with a point removed, and maps the complement of the neighbourhood to said point.
From both the trivialisation of the tangent bundles of 3-manifolds and the Thom-Pontryagin construction we now have a one-to-one correspondence between homotopy classes (cobordism classes) of the following:

$$\{\text{2-plane fields}\} \leftrightarrow \{\text{Maps } f : M \to S^2\} \leftrightarrow \{\text{Framed links}\}.$$

This lets us interchange between the three objects with plenty of freedom. An example of this is the obstruction invariants, we define below, which are invariants of 2-plane fields but are defined purely via the corresponding links.

5.2 Obstruction Invariants

These invariants of 2-plane fields will be our tool for proving the rest of our important theorems, and are particularly pertinent at deciding when 2-plane fields are ‘equivalent’, where by equivalent we really mean *homotopic* (as plane fields).

Definition 5.6. Let ξ, ξ' be 2-plane fields on M. We say ξ is homotopic to ξ', written $\xi \simeq \xi'$, if there exists a smooth family of 2-plane fields $\{\xi_t\}, t \in [0, 1]$, such that,

$$\xi = \xi_0 \text{ at } t = 0 \text{ and } \xi' = \xi_1 \text{ at } t = 1.$$

Informally, homotopy demands that we can continuously deform ξ into ξ'. We now remind ourselves of the correspondence between plane fields and framed cobordism classes of links.
The cobordism classes of links in our manifold that we discussed in the Thom-Pontryagin construction represent elements in the first homology group of our manifold, $H_1(M; \mathbb{Z})$ [2], where representing the same element in $H_1(M; \mathbb{Z})$, corresponds to being equivalent in the equivalence relation of *cobordism* i.e. being homologous.

Theorem 5.7. *[17] Let M be a closed, oriented 3-manifold. Any homology class $e \in H_1(M; \mathbb{Z})$ is represented by a link $L_e \subseteq M$. Two links L_1, L_2 are homologous, $[L_1] = [L_2] \in H_1(M; \mathbb{Z})$, if and only if L_1 and L_2 are cobordant.*

However, in the Thom-Pontryagin construction, we demanded our links to be *framed* and discussed *framed* cobordism classes of links. Unfortunately, when $[L_1] = [L_2] \in H_1(M; \mathbb{Z})$ and thus the links L_1, L_2 are cobordant, the framing of the two individual links does not necessarily extend over the cobordism between L_1 and L_2. To rectify this, we remove a disc D^2 from the surface $S \subseteq [0, 1] \times M$ so that the framing of L_1, L_2 extends over $S \setminus D^2 \subseteq [0, 1] \times M$ since this deformation retracts to a 1 dimensional complex of the two links, over which the 2-plane bundle is trivial [17]. We then attach a cylinder $[0, 1] \times S^1$ to our surface at the removed disc, such that:

$$\begin{aligned} &[0, 1] \times S^1 \cap \{0\} \times M = \emptyset, \\ &[0, 1] \times S^1 \cap \{1\} \times M = \{1\} \times S^1, \\ &[0, 1] \times S^1 \cap (S \setminus Int(D^2)) = \{0\} \times S^1 = \partial D^2. \end{aligned} \tag{5.2}$$

[2]refer to the Appendix for a discussion on Homology and Co-homology groups.

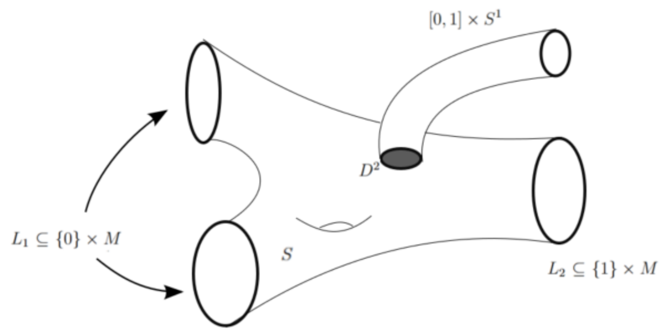

Figure 10: Extending the framing of L_1, L_2 over the cobordism.

Let us write (S^1, n) for a contractible loop (deformation retracts to a single point) with framing $n \in \mathbb{Z}$ (by which we mean if one were to take a second copy of the S^1 and push it in the direction of the framing of the first S^1, these loops would have linking number n). In such a case, we can attach (S^1, n) disjointly to another link to produce $L = L' \uplus (S^1, n)$. Subsequently, since the link $\{1\} \times S^1$ is not linked with any component of L_2, we identify this with (S^1, n). Thus, showing that L_1 is framed cobordant to $L_2 \uplus (S^1, n)$ in M. A description of which can be seen in Figure 10. It is worth noting the attached cylinder can be contained in a 3-ball and this is purely a trick necessary for the proof of Proposition 5.9.
Let $[L_\xi] \in H_1(M; \mathbb{Z})$ be the homology class of the link associated with ξ, then we may define the first of two invariants; the 'two dimensional' invariant.

Definition 5.8. [18] For 2-plane fields ξ_1, ξ_2 on M,

$$d^2(\xi_1, \xi_2) = PD[L_{\xi_1}] - PD[L_{\xi_2}] \in H^2(M; \mathbb{Z}),$$

where PD denotes the Poincaré dual [20] of an element in the homology group $H_1(M; \mathbb{Z})$.

Proposition 5.9. *[18] Let ξ_1, ξ_2 be 2-plane fields on M. Then, ξ_1 and ξ_2 are homotopic as plane fields on $M \setminus B^3$ [3] if and only if $d^2(\xi_1, \xi_2) = 0$.*

Proof. Suppose $d^2(\xi_1, \xi_2) = 0$ then $PD[L_{\xi_1}] = PD[L_{\xi_2}] \implies [L_{\xi_1}] = [L_{\xi_2}]$ and hence by Theorem 5.7 $\exists$ a surface $S \subseteq M \times [0, 1]$ such that,

$$\partial S = L_{\xi_1} \uplus L_{\xi_2} \subseteq M \times \{0\} \uplus M \times \{1\}.$$

We wish to extend the framing over the cobordism, which we can do, as discussed above, by removing a disk and attaching a cylinder. We form this new *framed* surface $S' \subseteq M \times [0, 1]$ with,

$$\partial S' = L_{\xi_1} \uplus (L_{\xi_2} \uplus (S^1, n)) \subseteq M \times \{0\} \uplus M \times \{1\}$$

so that L_{ξ_1} is framed cobordant to $L_{\xi_2'} = L_{\xi_2} \uplus (S^1, n)$, hence $\xi_1 \simeq \xi_2'$. After homotopy, ξ_2' and ξ_2 differ only by a contractible loop which is contained in a 3-ball, B^3, hence, agree away from this. By transitivity of homotopy, $\xi_1 \simeq \xi_2$ on $M \setminus B^3$. Conversely, if $\xi_1 \simeq \xi_2$ on $M \setminus B^3$ then ξ_1 and ξ_2 are homologous and hence their associated framed links live in the same homology class, that is $[L_{\xi_1}] = [L_{\xi_2}] \in H_1(M; \mathbb{Z})$. Thus, $PD[L_{\xi_1}] - PD[L_{\xi_2}] = 0 = d^2(\xi_1, \xi_2)$. □

[3] Note that saying plane fields are homotopic on the compliment of a ball is equivalent to being homologous, i.e. $d^2(\xi, \xi') \iff \xi, \xi'$ are homologous.

Given $d^2(\xi_1, \xi_2) = 0$, then $\xi_1 \simeq \xi_2$ on $M \backslash B^3$, but we would like to extend this homotopy over all of M, for which we must define another invariant. Thus, we introduce $d^3(\xi_1, \xi_2)$ to be the Hopf invariant of the map $f : S^3 \to S^2$, denoted $H(f)$. This is regarded as the 'three-dimensional' invariant contrary to its two-dimensional counterpart d^2.
We construct this map as $f_{\xi_1} \circ \pi_+$ on the upper hemisphere of S^3 and $f_{\xi_2} \circ \pi_-$ on the lower hemisphere, where π_+, π_- are the orthogonal projections of the respective hemispheres onto the equatorial disc of S^3. This is identified with the B^3 that contains the contractible loop and on which the homotopy, $\xi_1 \simeq \xi_2$, does not extend. The Hopf invariant, $H(f)$, is also defined as the linking number of the pre-images of two regular values of the map. If $y, y' \in S^2$, and we have a map $f : M \to S^2$ then $f^{-1}(y), f^{-1}(y')$ are links in M and the Hopf invariant of f is defined as their linking number, taking integer value. This invariant in fact gives an isomorphism $\pi_3(S^2) \xrightarrow{\sim} \mathbb{Z}$. It is clear that both invariants are transitive, that is

$$d^i(\xi_1, \xi_3) = d^i(\xi_1, \xi_2) + d^i(\xi_2, \xi_3), i = 2, 3.$$

Thus, the d^3 invariant extends the homotopy of plane fields over the full manifold. Hence ξ_1, ξ_2 are homotopic if and only if they have equivalent d^2 and d^3 values [17]. Thus the homotopy classes of 2-plane fields are completely described by a pair (d^2, d^3), which every plane field in the homotopy class possesses.

5.3 Lutz Twisting... Again

We have now built up some heavier machinery regarding 2-plane fields, namely the correspondences between 2-plane fields, maps $M \to S^2$ and framed links, along with our obstruction invariants. We will now provide further detail on Lutz twists and their effect on d^2 and d^3.
The set up is as before, we obtain ξ' from ξ via a Lutz twist, along a transverse knot $K \subseteq (M, \xi)$, whose tubular neighbourhood we identify with $S^1 \times D^2$, in which K is identified with $S^1 \times \{\mathbf{0}\}$. The respective contact structures are given by the kernels of,

$$\alpha = d\theta + r^2 d\phi \text{ and } \alpha' = f_1(r)d\theta + f_2(r)d\phi.$$

It will be useful to understand the maps $f_\xi, f_{\xi'} : M \to S^2$ associated to the respective plane fields. We can define f_ξ on $S^1 \times D^2 \subseteq M$, with coordinates $(\theta, (x, y)) \in S^1 \times D^2$ and can trivialise $T(S^1 \times D^2)$ with orthogonal basis $\{\partial_\theta, \partial_x, \partial_y\}$. Then for ξ', a vector normal to ξ' is given by $f_1(r)\partial_\theta + f_2(r)\partial_\phi$, thus the Gauss map $f_{\xi'}$ sends points to this normal vector. We see $(1, 0, 0) \in S^2$, which corresponds to ∂_θ, is the only non-regular value for the map $f_{\xi'} : M \to S^2$, as it is obtained at $\{r = r_0\}$ determined by $f_2(r_0) = 0$. In such a case, the image of $f_{\xi'}$ will be $f_1(r)\partial_\theta$ and hence its derivative will not be surjective.
Since the point $(-1, 0, 0) \in S^2$ is regular, then its pre-image under $f_{\xi'}$ is a link in $M = S^1 \times D^2$. With the usual choice of orientations, $\partial\theta$ corresponding to the positive direction of $S^1 \times \{\mathbf{0}\} \subseteq S^1 \times D^2$, we have that $f_{\xi'}^{-1}(-1, 0, 0) = -S^1 \times \{0\}$. More generally, for $p \in S^2 \setminus \{(1, 0, 0)\}$ the pre-image $f_{\xi'}^{-1}(p)$ is a circle $S^1 \times \{\mathbf{u}\}, \mathbf{u} \in D^2$, with orientation given by ∂_θ (valid only in the domain $\{(\theta, r, \phi) : f_2(r) < 0\} = \{r < r_0\}$) [17].

Proposition 5.10. *[18] Let ξ, ξ' be contact structures on a 3-manifold, M and let K be a positively transverse knot in (M, ξ). If ξ' is obtained from ξ by a Lutz twist along K, then $d^2(\xi, \xi') = PD[K]$.*

Proof. Consider the contact structure ξ on M given by

$$\xi = ker(d\theta + r^2 d\phi) = ker(d\theta + xdy - ydx).$$

We identify a neighbourhood of K with $S^1 \times D^2$ and give it a framing of $\{x_1, x_2, x_3\}$ with $x_1, x_2 \in \xi$ and $x_3 \in \xi^\perp$. That is, x_1, x_2 are tangent to ξ and x_3 is normal to ξ. Then the Gauss map $f_\xi : M \to S^2$

has constant image $(0,0,1) \in S^2$. We then change the trivialisation of TM on $S^1 \times D^2$, such that we now have an oriented basis $\{\partial_\theta, \partial_x, \partial_y\}$. The Gauss map f_ξ is still sending points onto vectors normal to ξ, however is no longer constant. Importantly, we still have $f_\xi(p) \neq -\partial_\theta$ which is identified with $(-1,0,0) \in S^2$. Since $(-1,0,0)$ is not in the image of f_ξ, then $L_\xi = \emptyset$, hence $PD[L_\xi] = 0$. When we Lutz twist along K, we replace the contact structure on $S^1 \times D^2$ with

$$\xi' = ker\left(f_1(r)d\theta + f_2(r)d\phi\right).$$

From the discussion above, at $r = 0$, the contact form is spanned by $-\partial_\theta$ i.e. along K in the negative direction, since we identify K with $S^1 \times \{0\} \subseteq S^1 \times D^2$. Hence, $f_{\xi'}^{-1}(-1,0,0) = -S^1 \times \{0\} = -K$ so $L_{\xi'} = -K$. Thus, we achieve

$$d^2(\xi, \xi') = 0 - PD[L_{\xi'}] = PD[K].$$

□

6 Eliashberg's Classification of Overtwisted Contact Structures

In Section 4.1 we discussed the dichotomy between overtwisted and tight contact structures. In 1989, Russian mathematician Yakov Eliashberg set about trying to classify contact structures, in particular those that are overtwisted [5]. It initially seemed that classifying overtwisted contact structures would be a far more complicated task than classifying their tight counterparts, however this turns out to be precisely the opposite of reality. The classification of overtwisted contact structures reduces to 'simple' homotopy theory of 2-plane fields, whilst classifying tight contact structures is difficult even for simple manifolds.

Fix a point $p \in M$ and an embedded (overtwisted) disc $\Delta \subseteq M$ centred at the fixed point $p \in M$. Denote the space of all contact structures that are overtwisted along the disk Δ by $\mathcal{H}_{ot}$. These overtwisted contact structures certainly form a subset of all 2-plane fields tangent to Δ on M; this is just a restriction on integrability. The space of 2-plane fields tangent to Δ is denoted, $\mathcal{H}_{pf}$. Thus, we have a natural inclusion, $i : \mathcal{H}_{ot} \hookrightarrow \mathcal{H}_{pf}$. The statement of Eliasberg's theorem is as follows.

Theorem 6.1 (Eliashberg)**.** *[5] The inclusion*

$$i : \mathcal{H}_{ot} \hookrightarrow \mathcal{H}_{pf} \tag{6.1}$$

is a homotopy equivalence.

Recall, a homotopy equivalence induces isomorphisms between all homotopy, homology and cohomology groups on the two spaces. This is a tremendously powerful theory as it lets us decide if two overtwisted contact structures are the same (up to isotopy) purely via homotopy theory of plane fields, as discussed above. Essentially, two overtwisted contact structures are isotopic if they are homotopic as plane fields. The classification also states that there must be an overtwisted contact structure in *every* homotopy class of plane fields.

Theorem 6.2. *[18] Let ξ be a 2-plane field on M. Then there exists an overtwisted contact structure homotopic to ξ, as a 2-plane field.*

Proof. We can find a positively transverse knot K such that $d^2(\xi_0, \xi) = PD(K)$. We can then perform a Lutz twist in ξ_0 along K to get a contact structure ξ_1, by Proposition 5.10, $d^2(\xi_0, \xi_1) = PD(K)$. By transitivity, we have,

$$d^2(\xi_1, \xi) = d^2(\xi_0, \xi) - d^2(\xi_0, \xi_1) = PD(K) - PD(K) = 0.$$

Hence, $\xi_1 \simeq \xi$ on the compliment of a 3-ball in M, i.e. on $M \setminus B^3$. Consider a transverse knot, $T \subseteq (B^3, \xi_1)$, then T is null-homologous in the homology of the manifold, thus $PD(T) = 0 \in H^2(M; \mathbb{Z})$. Let us perform a Lutz twist of ξ_1 along T, resulting in ξ_2. Then,

$$d^2(\xi_2, \xi) = d^2(\xi_2, \xi_1) + d^2(\xi_1, \xi) = PD(T) + 0 = 0.$$

Let us trivialise TB^3 with the basis as seen in the proof of Proposition 5.10, that is we have $\{x_1, x_2, x_3\}$ with $x_1, x_2 \in \xi_0$ and $x_3 \in \xi^\perp$ on $\xi_0|_{B^3} = \xi_1|_{B^3}$. We now wish to compute the $d^3(\xi_2, \xi_1)$, which is the linking number of two pre-images of the map $f : S^3 \to S^2$, determined by $f_{\xi_2}|_{B^3}$ on the upper hemisphere and $f_{\xi_1}|_{B^3}$ on the lower hemisphere. The map $f_{\xi_1}|_{B^3}$ avoids a neighbourhood $U \subset S^2$ of the point $(-1, 0, 0)$ as this is not in the image of f_{ξ_1}. We also note that since ξ_2 is obtained from ξ_1 via a Lutz twist along T, from the discussion at the beginning of Section 5.3, we have $f^{-1}(-1, 0, 0) = -T$. Then $d^3(\xi_2, \xi_1)$ is the linking number of $f^{-1}(-1, 0, 0) = -T$ and $f^{-1}(u)$ for a regular value $u \in U \setminus \{(-1, 0, 0)\}$. Again from Section 5.3, $f^{-1}(u)$ will be the circle $S^1 \times \{\mathbf{a}\}$ in the knot neighbourhood of $-T$, also known as a *transverse push-off* of $-T$. Hence, $d^3(\xi_2, \xi_1)$, defined as the linking number of $f^{-1}(-1, 0, 0)$ and $f^{-1}(u)$, will be the self-linking number of T, that is $d^3(\xi_2, \xi_1) = sl(T)$. By additivity we have,

$$d^3(\xi_2, \xi) = d^3(\xi_2, \xi_1) + d^3(\xi_1, \xi) = sl(T) + d^3(\xi_1, \xi).$$

Hence, with a suitable choice of $T \subseteq (B^3, \xi_1)$ [4] we can have $d^3(\xi_2, \xi_1) = 0$ and hence ξ_2 is homotopic to ξ, that is $\xi_2 \simeq \xi$ where ξ_2 is overtwisted by Corollary 4.11. Thus there is an overtwisted contact structure in each homotopy class. □

This forms the surjectivity part of Eliashberg's theorem, that there is a unique overtwisted contact structure in each homotopy class. It will assist in the proof of Theorem 7.5.

7 Open Books Supporting Overtwisted Contact Structures

In Sections 3 and 4 we defined the notion of two inherently different mathematical objects one can impose on a manifold, however the Giroux correspondence miraculously puts these in a one-to-one correspondence with one another. Using only a small part of this correspondence, we want to understand what the open books which support overtwisted contact structures look like. Are there any restrictions on the type of open book or is it a completely arbitrary assignment? We first need to think about what it means for an open book to *support* a contact structure.

Definition 7.1. Let (M, ξ) be a contact 3-manifold. The contact structure ξ is *supported* by an open book decomposition (B, π) on M, if there exists a contact 1-form α such that, $d\alpha$ is a positive area form on each page Σ_θ of the open book and $\alpha > 0$ on B. In such a case we write $\xi_{(B,\pi)}$ when ξ is being supported by the OBD (B, π).

What we want to have is that our contact structure ξ can be isotoped arbitrary close to being tangent to the pages of the open book away from the binding. Before twisting as one approaches the binding, and being transverse to some tubular neighbourhood of the binding. One can think of the contact structure starting off tangent to a page far away from the binding and slowly twisting as it approaches the binding, so that it has made a $\pi/2$ twist by the time it is at the binding, before continuing through the other side, completing a full π twist to be tangent to the antipodal page.

Example 7.2. [17] [21] We now wish to extend Example 3.3, to show that the open books (U, π) and (H^+, π_+) both support the standard contact structure on S^3. We consider S^3 as the unit sphere in $\mathbb{C}^2$ with the contact structure given by the kernel of $\alpha = r_1^2 d\theta_1 + r_2^2 d\theta_2$.

[4] Refer to [18] for a thorough justification and explanation of this. It goes into detail explaining how to compute self-linking numbers as the *writhe* of a front projection and justifies $d^3(\xi_2, \xi_1) = 0$.

1. For (U, π_U) the binding is given by $U = \{z_1 = 0\}$ and hence $\alpha = r_2^2 d\theta_2$ on U, which certainly satisfies $\alpha > 0$. Using the properties from Proposition 2.16, we have

$$d\alpha = 2dr_1 \wedge d\theta_1 + 2dr_2 \wedge d\theta_2, \tag{7.1}$$

restricting to $2r_2 dr_2 \wedge d\theta_2$ on the pages, which is a positive area form for each θ. Therefore, (U, π_U) supports the standard contact structure on S^3.

2. Similarly for (H^+, π_+), on the components of the binding, $\{r_1 = 0\}$ and $\{r_2 = 0\}$, we have respective tangential vectors $\partial_{\theta 1}$ and $\partial_{\theta 2}$ which satisfy $\alpha(\partial_{\theta 1}) = \alpha(\partial_{\theta 2}) = 1 > 0$. Then, it is possible to re-write the parametrisation of the pages in polar coordinates as the map

$$f(r, \theta) = (\sqrt{t}, \theta, \sqrt{1-t}, \phi - \theta), \tag{7.2}$$

with $t \in (0,1), \theta \in S^1$. We can then pullback α by f to get the form on the annulus $(0,1) \times S^1$:

$$f^*\alpha = (\sqrt{t})^2 d\theta + (1-t) + (\sqrt{1-t})^2 d(\phi - \theta) = td\theta + (1-t)(-d\theta) = (2t-1)d\theta. \tag{7.3}$$

This is of course a positive area form on the annulus $(0,1) \times S^1$. Subsequently, α is a positive area form on the pages of (H^+, π_+) and hence supports the standard contact structure on S^3.

We can also conduct similar working to show that (H^-, π_-) supports the standard overtwisted contact structure on S^3; a proof of which can be found in [22].

Proposition 7.3. *[22] The Contact structure supported by the negative Hopf link, $\xi_{(H^-, \pi_-)}$, is overtwisted.*

It should not be a surprise to find that every open book supports a contact structure:

Theorem 7.4 (Thurston-Winkelnkemper)**.** *[23] Every open book decomposition (Σ, ϕ) supports a contact structure ξ_ϕ on M_ϕ.*

For the sake of completeness, this provides a proof of the existence of contact structures on 3-manifolds when used in conjunction with Theorem 3.2.
We now turn our attention to our final theorem, regarding the nature of the Giroux correspondence and specifically the relationship between overtwisted contact structures and negatively stabilised OBDs.

Theorem 7.5. *A contact structure is overtwisted if and only if it is supported by an open book with a negative stabilisation.*

Before we prove this theorem, we have some work to do regarding the nature of open books supporting contact structures. Below we state some facts regarding connect sums of contact structures in preparation for a proof of Theorem 7.5.

Definition 7.6 (Contact connected sum)**.** Let $(M_0, \xi_0), (M_1, \xi_1)$ be contact manifolds. Then we can form their *contact connected sum*, $(M_0 \# M_1, \xi_0 \# \xi_1)$, via the orientation reversing diffeomorphism

$$f : \partial(\overline{M_0 \setminus B_0}) \to \partial(\overline{M_1 \setminus B_1}), \tag{7.4}$$

where B_0, B_1 are balls in the respective manifolds. $M_0 \# M_1$ is formed by gluing $M_0 \setminus B_0$ to $M_1 \setminus B_1$ by f such that we also preserve the contact structure and

$$\xi_0|_{M_0 \setminus B_0} \cup \xi_1|_{M_1 \setminus B_1},$$

extends to a well-defined contact structure on $M_0 \# M_1$.

The balls we can choose to be arbitrarily small such that when one glues the contact structures, we can invoke the Darboux's theorem on the local structure of contact structures to get a well defined contact structure around the gluing.

Proposition 7.7. *Let $\xi_{(\Sigma_0,\phi_0)}, \xi_{(\Sigma_1,\phi_1)}$ be contact structures on M supported by the open books, $(\Sigma_0,\phi_0),(\Sigma_1,\phi_1)$. Then,*

$$\xi_{(\Sigma_0,\phi_0)} \# \xi_{(\Sigma_1,\phi_1)} = \xi_{(\Sigma_0,\phi_0)*(\Sigma_1,\phi_1)}$$

Proof. This follows immediately from Theorem 3.12 as both contact structures are already well defined on M. □

It will also prove useful to understand the affect of stabilisation on our obstruction invariants d^2 and d^3. When one stabilises an open book, the one-handle attached to the page can be completely contained in a 3-ball and hence the open book remains unchanged on the compliment of this 3-ball. Since we say contact structures are homologous if they are homotopic on the compliment of a 3-ball, stabilising will not affect the homology class of the associated contact structures. A discussion and proof of this statement can be found in [22]. Hence, the d^2 invariant remains unchanged under stabilisations. However, the same is not true for our three dimensional invariant, d^3. This begs the question; what is the effect of stabilisation on d^3? We first remind ourselves that $d^3 \in \mathbb{Z}$, since it is defined as a linking number of two links. We summarise the affect of negative stabilisation on d^3 in the below Proposition 7.8, which does much of the heavy lifting in the proof of Theorem 7.5.

Proposition 7.8. *[22] Let ξ' be the negative stabilisation of ξ, then $d^3(\xi,\xi') = +1$.*

Proof. Refer to [22] for a proof by Giroux and Goodman. □

Proof of Theorem 7.5, [24]. Suppose ξ is supported by a negatively stabilised OBD (Σ,ϕ), then $(\Sigma,\phi) = (\Sigma',\phi') * (H^-,\pi_-)$, for some arbitrary OBD (Σ',ϕ') . Thus we have,

$$\xi_{(\Sigma,\phi)} = \xi_{(\Sigma',\phi')*(H^-,\pi_-)} = \xi_{(\Sigma',\phi')} \# \xi_{(H^-,\pi_-)}.$$

We know the contact structure associated to the negative Hopf link is overtwisted by Proposition 7.3. Thus, if we glue the contact structures away from some overtwisted disk Δ (which is allowed as the connect sum is well-defined no matter of the choice of balls $B_i \subseteq M_i$), then $\xi_{(\Sigma,\phi)}$ also contains Δ, hence is overtwisted. Conversely, suppose ξ is overtwisted, supported by some OBD (Σ,ϕ) and lives in the homotopy class with obstruction pair (d^2,d^3). Let ξ' be the plane field representing the homotopy class with same d^2 but d^3 shifted by minus one, that is the pair (d^2, d^3-1). By Theorem 6.2, there exists an overtwisted contact structure homotopic to ξ' and thus there is a contact structure in this homotopy class. Let ξ' represent this contact structure. Then it is supported by an open book (Σ',ϕ') by Theorem 7.4. We then negatively stabilise this contact structure to produce $\eta = \xi_{(\Sigma',\phi')*(H^-,\pi_-)}$. As was discussed above, stabilising has no affect on d^2, but shifts d^3 by $+1$ by Proposition 7.8, hence η has obstruction pair (d^2,d^3) and $\eta \simeq \xi_{(\Sigma,\phi)}$, as plane fields. Since both η and $\xi_{(\Sigma,\phi)}$ are overtwisted, by Eliashberg's classification, they are isotopic. Hence, ξ is supported by an open book with a negative stabilisation. □

A Appendix

A.1 Homology and Co-Homology

For a thorough introduction to homology and co-homology, please refer to Alan Hatcher's bible of Algebraic Topology, [20]. It would take too long and too much of a divergence from contact topology to define homology and it is expected that the reader has some knowledge of homology and co-homology, hence we state only the relevant material.
For a smooth n-manifold, M, we have a remarkable property called the *Poincaré duality*, which defines an isomorphism between homology and co-homology groups of an n-manifold;

$$PD : H_k(M;\mathbb{Z}) \xrightarrow{\sim} H^{n-k}(M;\mathbb{Z})\,\forall k \tag{A.1}$$

This is applicable for the project as for our closed, oriented 3- manifolds, M, then $H_1(M;\mathbb{Z}) \cong H^2(M;\mathbb{Z})$. We use this in the discussion of 2-plane fields since the framed cobordism classes of links correspond to elements in $H_1(M;\mathbb{Z})$, as stated in 5.7.

This is an area of mathematics where it can be difficult to visualise the objects in question. I recommend "A visual introduction to the Giroux correspondence" on Patrick Massot's webpage for some excellent animations and photos of open books and contact structures. https://www.imo.universite-paris-saclay.fr/ patrick.massot/en/exposition/giroux_correspondance.html

References

[1] Horst E Winkelnkemper. Manifolds as open books. *Bulletin of the American Mathematical Society*, 79(1):45–51, 1973.

[2] James W Alexander. Note on riemann spaces. 1920.

[3] William C Hoffman. The visual cortex is a contact bundle. *Applied Mathematics and Computation*, 32(2-3):137–167, 1989.

[4] Emmanuel Giroux. Géométrie de contact: de la dimension trois vers les dimensions supérieures. *arXiv preprint math/0305129*, 2003.

[5] Yakov Eliashberg. Classification of overtwisted contact structures on 3-manifolds. *Inventiones mathematicae*, 98(3):623–637, 1989.

[6] Victor Guillemin and Alan Pollack. *Differential topology*, volume 370. American Mathematical Soc., 2010.

[7] Joel W Robbin and Dietmar A Salamon. Introduction to differential topology. *Book project*, 2011.

[8] John W Milnor and David W Weaver. *Topology from the differentiable viewpoint*, volume 21. Princeton university press, 1997.

[9] Anton Izosimov. *Lecture notes on Submanifolds*. University of Arizona, 2018.

[10] Nikolai Saveliev. *Lectures on the topology of 3-manifolds: an introduction to the Casson invariant*. Walter de Gruyter, 2011.

[11] John B Etnyre. Lectures on open book decompositions and contact structures. *arXiv preprint math/0409402*, 2004.

[12] Robert Myers. Open book decompositions of 3-manifolds. *Proceedings of the American Mathematical Society*, pages 397–402, 1978.

[13] Dale Rolfsen. *Knots and links*. Number 346. American Mathematical Soc., 2003.

[14] Steven Sivek. *Math 273: Contact Geometry in 3 Dimensions, Lecture 23*. Harvard University, 2012.

[15] Ryosuke Yamamoto. Open books supporting overtwisted contact structures and the stallings twist. *Journal of the Mathematical Society of Japan*, 59(3):751–761, 2007.

[16] John B Etnyre. Introductory lectures on contact geometry. *arXiv preprint math/0111118*, 2001.

[17] Hansjörg Geiges. An introduction to contact topology. *Cambridge Studies in Advanced Mathematics/Cambridge University Press*, 109, 2008.

[18] Steven Sivek. *Math 273: Contact Geometry in 3 Dimensions, Lecture 2*. Harvard University, 2012.

[19] Riccardo Benedetti and Paolo Lisca. Framing 3-manifolds with bare hands. *L'Enseignement Mathématique*, 64(3):395–413, 2019.

[20] Allen Hatcher. *Algebraic topology*. Cambridge Univ. Press, Cambridge, 2000.

[21] Steven Sivek. *Math 273: Contact Geometry in 3 Dimensions, Lecture 21*. Harvard University, 2012.

[22] Emmanuel Giroux and Noah Goodman. On the stable equivalence of open books in three-manifolds. *Geometry & Topology*, 10(1):97–114, 2006.

[23] William P Thurston and Horst E Winkelnkemper. On the existence of contact forms. *Proceedings of the American Mathematical Society*, pages 345–347, 1975.

[24] Noah Goodman. *Contact structures and open books*. The University of Texas at Austin, 2003.